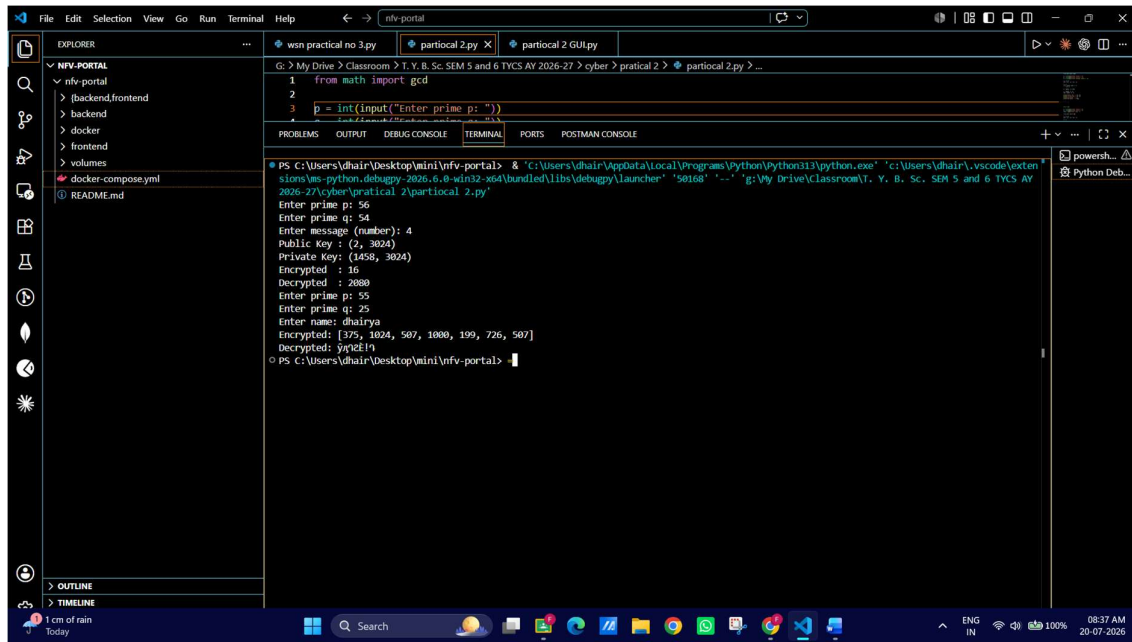


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: cyber security

Practical 2: Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

CLI OUTPUT

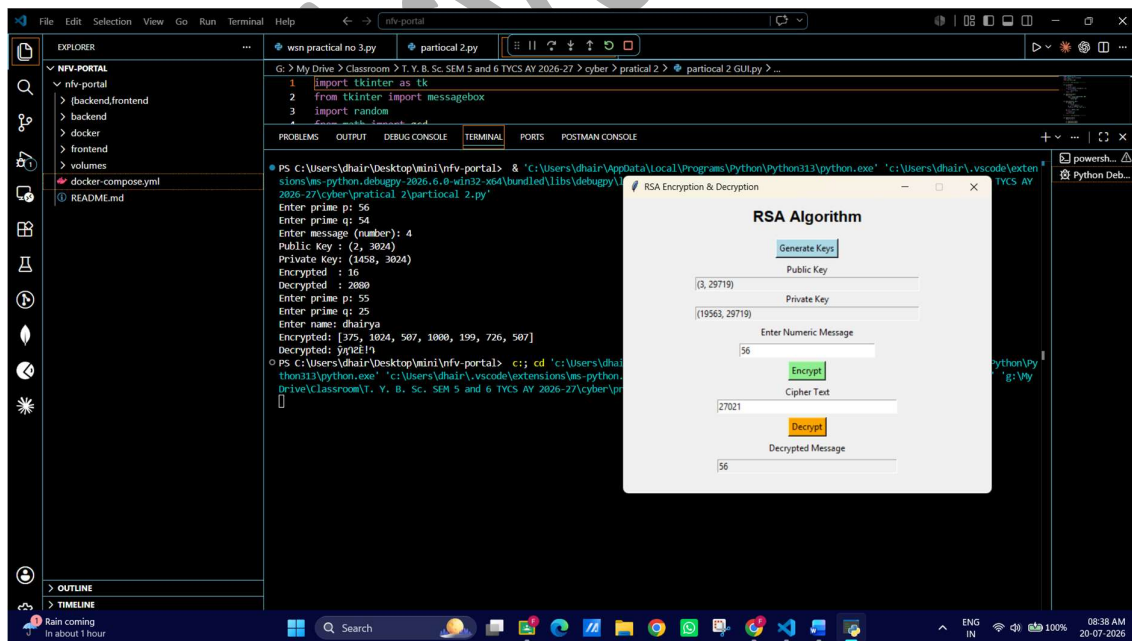


```
1 from math import gcd
2
3 p = int(input("Enter prime p: "))
4 q = int(input("Enter prime q: "))
5
6 # Generate Public and Private Keys
7 def generate_keys(p, q):
8     n = p * q
9     phi = (p - 1) * (q - 1)
10    e = 2
11    while gcd(e, phi) != 1:
12        e += 1
13    d = 1
14    while (e * d) % phi != 1:
15        d += 1
16    return (n, e, d)
17
18 # Encryption
19 def encrypt(message, n, e):
20     encrypted = ""
21     for char in message:
22         char_int = ord(char)
23         encrypted_int = (char_int ** e) % n
24         encrypted += str(encrypted_int) + " "
25     return encrypted
26
27 # Decryption
28 def decrypt(encrypted_message, n, d):
29     decrypted_message = ""
30     encrypted_ints = encrypted_message.split(" ")
31     for encrypted_int in encrypted_ints:
32         encrypted_int = int(encrypted_int)
33         decrypted_int = (encrypted_int ** d) % n
34         decrypted_message += chr(decrypted_int)
35     return decrypted_message
36
37 # Main
38 if __name__ == "__main__":
39     p = 2
40     q = 3024
41     n, e, d = generate_keys(p, q)
42     message = "dhairya"
43     encrypted_message = encrypt(message, n, e)
44     decrypted_message = decrypt(encrypted_message, n, d)
45     print("Encrypted: ", encrypted_message)
46     print("Decrypted: ", decrypted_message)
```

PS C:\Users\dhair\Deskto\mini\nvf-portal> & "C:\Users\dhair\AppData\Local\Programs\Python\Python313\python.exe" "C:\Users\dhair\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher" "58168" "-." "g:\Vy Drive\Classroom\T. Y. B. Sc. SEM 5 and 6 TYCS AY 2026-27\cyber\practical 2\partical 2.py"

Enter prime p: 56
Enter prime q: 54
Enter message (number): 4
Public Key : (2, 3024)
Private Key: (1458, 3024)
Encrypted : 16
Decrypted : 2080
Enter prime p: 55
Enter prime q: 25
Enter name: dhairya
Encrypted: [375, 1024, 507, 1080, 199, 726, 507]
Decrypted: dhairya
PS C:\Users\dhair\Deskto\mini\nvf-portal>

GUI OUTPUT:



Name: Dhairya Singh
Roll no: T119